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DEPARTMENT OF PEDIATRICS AND CHILD HEALTH

Neonatal Sepsis

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Outline

- ▶ Terminologies
- ▶ Introduction
- ▶ Epidemiology and Pathogenesis
- ▶ Etiology
- ▶ Clinical presentations
- ▶ Diagnosis and management
- ▶ References



Objectives

- ▶ Review of terminologies associated with neonatal infections
- ▶ Review risk factors and pathophysiology of neonatal infections
- ▶ Understand presentations of neonatal sepsis
- ▶ Review most common organisms and treatments of neonatal sepsis



Terminologies

- ▶ **Neonatal period**-is the first 28 day of life, and also classified as
 1. Early neonatal period (the *first 7 days*)
 2. Late neonatal period (from *day 8 to 28 days*)
- ▶ **Perinatal period**- the period from 28 weeks of gestation to 7 days postnatal
- ▶ **Prenatal period**- the period from 28 weeks of gestation to birth
- ▶ **Postnatal period**- the period after birth



Introduction

- ▶ Infections in the newborn are often classified by *their timing* relative to birth and include
 - ▶ Congenital
 - ▶ Perinatal
 - ▶ Early-onset, and
 - ▶ Late-onset disease.
- ▶ **Congenital** or **intrauterine** infections.....*are those transmitted across the placenta*
- ▶ **perinatal** infection those transmitted from the mother to the *fetus* or *newborn* infant during the birth process.



Unique characteristics of neonatal infections

- Transmission from the mother occurs via diverse modes
- Maternal infections are often not diagnosed during pregnancy (asymptomatic, non-specific)
- Etiologies are different (bacteria, fungi, virus, protozoa)
- Immature immunity
- Comorbid conditions complicate Dx & Mx of infections
- c/ms are variable (subclinical, mild to severe, focal/systemic, malformations) depending on :
 - Timing of exposure
 - Inoculum size
 - Immune status
 - Virulence of organisms



WHY All neonates are susceptible to infection?

- **The immaturity of a neonate's immune system** makes them MORE SUSCEPTIBLE to sepsis.

Immunologic deficiencies



- ▶ The immune system can be divided into 4 different parts:
 1. **Antibodies (immunoglobulins)** - which damage organisms and attract phagocytic cells.
 2. **Lymphocytes** - produce antibodies & kill organisms.
 3. **Phagocytic cells** – (macrophages and polymorphs), that ingest and kill organisms.
 4. **Complement** - a group of proteins that help antibodies to damage organisms and attract phagocytic cells.



WHY DO NEWBORN INFANTS OFTEN BECOME INFECTED?

Antibodies

1. The antibodies *IgA and IgM* are too big to cross the placenta from the mother. But, **during the first 6 months of life the infant gradually produces its own IgA and IgM.**
2. IgG does cross the placenta but mostly crosses in the last weeks of pregnancy. Preterm infants, therefore, have very little IgG. After birth the infant starts to produce its own IgG as the amount of maternal IgG decreases.

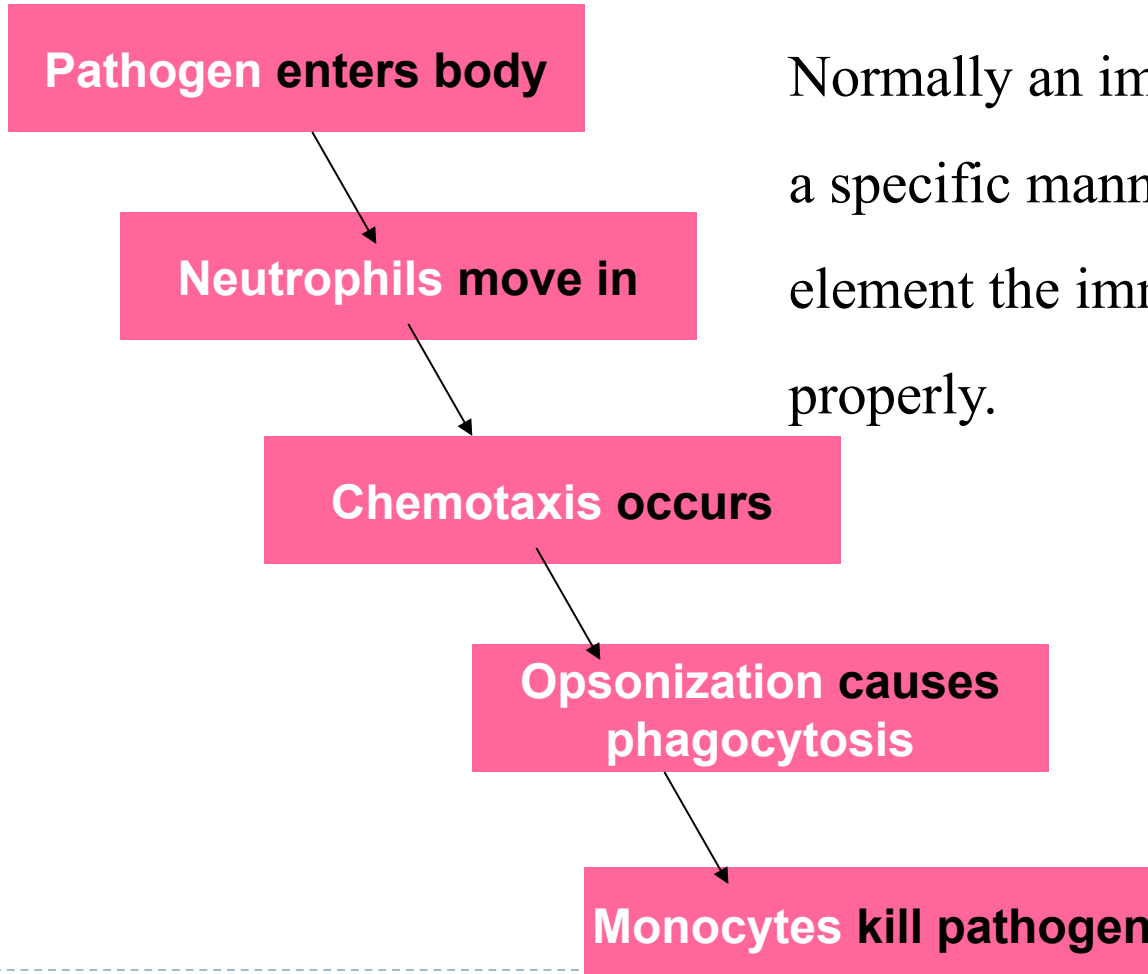
WHY DO NEWBORN INFANTS OFTEN BECOME INFECTED?



Lymphocytes, Phagocytic cells, Complements

3. All infants have immature lymphocytes. A mature lymphocyte needs previous contact with a specific organism before it is able to recognise and kill it.
4. The phagocytic cells are present in all infants but they do not function normally due to the low concentrations of complement.
5. Low complement concentrations in (especially if preterm) prevents the phagocytes from functioning normally.

What makes a neonate's immune system immature?



Normally an immune system responds to a pathogen in a specific manner, but if there are problems with any element the immune system is unable to function properly.

newborn to infection

- **Related to immaturity of both the cellular and humoral immune systems at birth.**
 - ❖ Decreased function of neutrophils
 - ❖ Neonatal T cells appear to be relatively deficient in most of their major function
 - ❖ Low concentrations of immunoglobulin (preterm)

Conti....

- ❖ Quantitative and qualitative defects of complement system (both term & preterm)
- ❖ In premature newborns cord IgG levels are directly proportional to GA.
- ❖ **IgA is virtually absent at birth (usually <5mg/dl)**
- ❖ Newborn usually lack antibody mediated protection **against Escherichia coli & other Enterobacteriaceae. (Because no IgM is transferred to fetus)**



General Approach

- ▶ The general approach includes a *detailed maternal history*
- ▶ Comprehensive physical examination.
- ▶ **laboratory testing** to establish a *specific microbiologic etiology* and *guide management decisions*.



Epidemiology & pathogenesis

- ▶ Important causes of neonatal morbidity & mortality.
- ▶ 2% of fetuses are infected in utero, and
- ▶ up to 10% of infants have infections in the 1st mo of life.
- ▶ **ROUTES OF INFECTION**
 - ▶ 1. Transplacentally
 - 2. Ascending Infection/ Intrapartum
 - 3. Postnatally



Pathogenesis of ascending infections

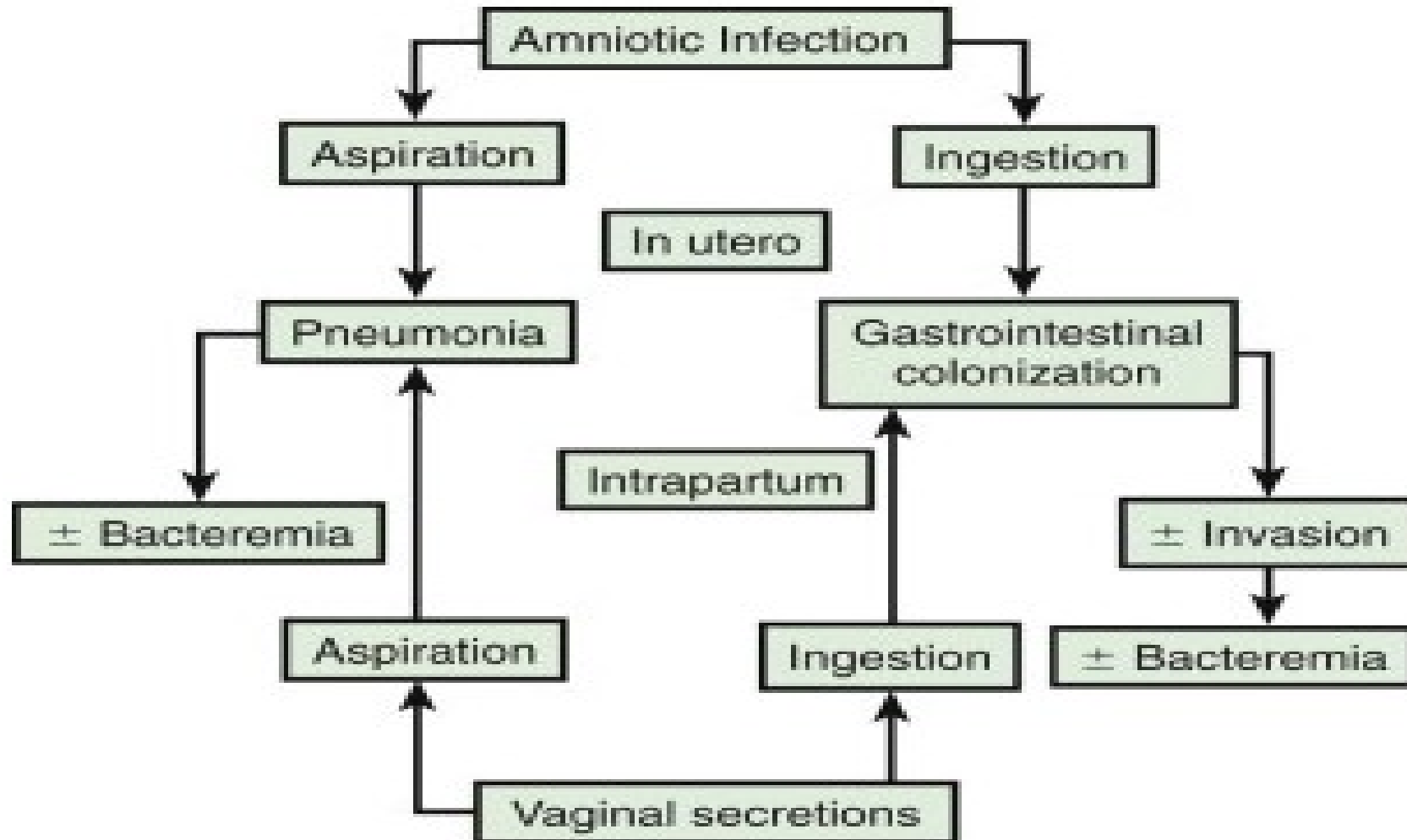
- ▶ The human birth canal is colonized with **aerobic** and **anaerobic organisms** may result in **ascending amniotic infection** and/or **colonization of the neonate at birth**
- ▶ Bacterial vertical transmission occurs while in utero or more commonly ***during labour or delivery.***



Conti....

- ▶ Bacterial colonization **does not always** result in disease.
- ▶ In the absence of Intrapartum antibiotics prophylaxis, approximately **50%** of infants born to mothers colonized with GBS are found to be colonized with this organism at birth.
- ▶ Approximately **1% to 2% of all colonized** infants develop **invasive GBS disease.**

Pathogenesis of ascending infections





Conti...

■ Factors that **increase risk of infection**:

- ▶ Prolonged ROM (>18hrs)
- ▶ Untreated maternal urinary/GI infection
- ▶ Instrumental delivery
- ▶ Umbilical catheterization
- ▶ Prolonged labor

■ Factors that **increase development of disease** from infection:

- Prematurity
- Low birth-weight
- Underlying illnesses
- Inoculum size
- Virulence of the organism
- the innate immune system,
- host response, and transplacental maternal antibodies .



Conti.....

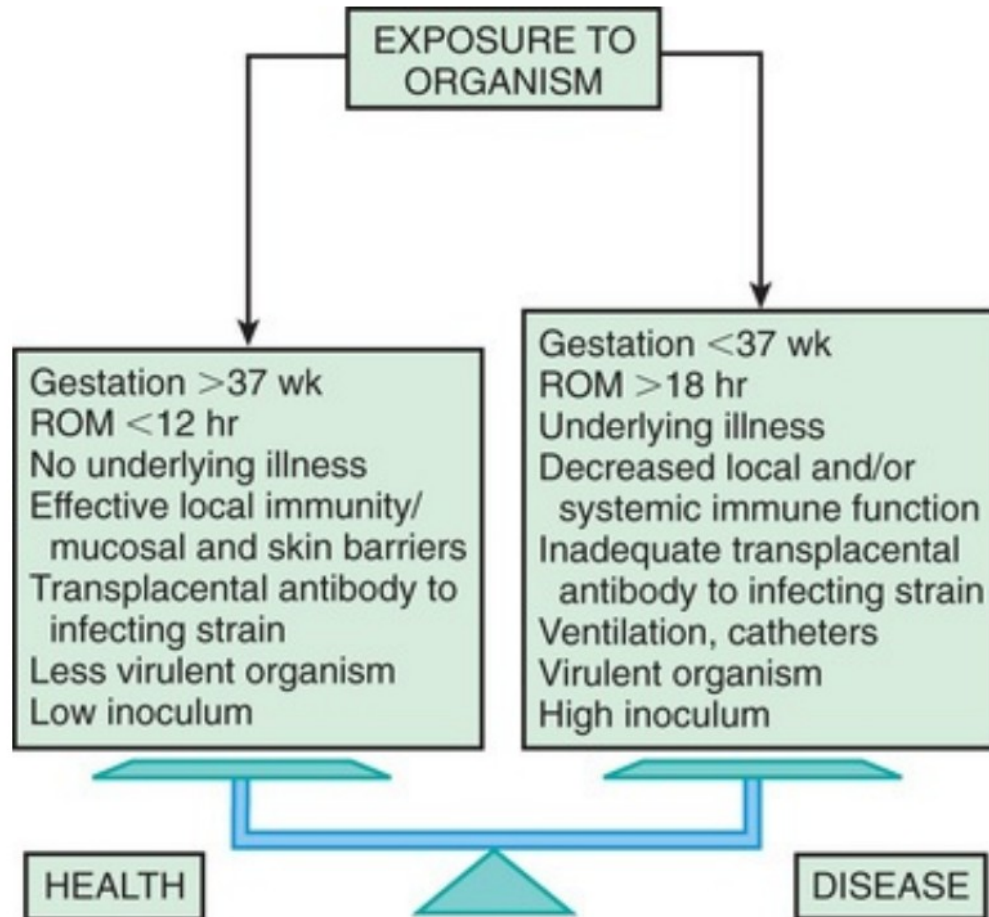


FIG. 129.5 Factors influencing the balance between health and disease in neonates exposed to a potential pathogen. ROM, Rupture of membranes. (Adapted from Baker CJ: Group B streptococcal infections, *Clin Perinatol* 24:59–70, 1997.)

- ▶ Aspiration of infected amniotic fluid leads to **congenital pneumonia** or systemic infection manifesting at:-
 - Before delivery (fetal distress, tachycardia)
 - At delivery (PNA)
 - After delivery (respiratory distress, shock)
- ▶ Colonization during labour or delivery leads to infection within **1-2 days** after exposure



Etiology of perinatal infections

- ▶ Any microorganism inhabiting the GUT or lower GIT may cause intrapartum and postpartum infection.
- ▶ **Most common bacteria**:-GBS, enteric organisms(E.coli,Kleibsell), gonococci, and chlamydia
- ▶ **Common viruses**:-CMV, HSV, enteroviruses, and HIV.



Cont'd...

- ▶ Etiology for **Pneumonia**:-GBS, gram-negative enteric aerobes, *L. monocytogenes*, *Mycoplasma spp*, *C. trachomatis*, CMV, HSV, and *Candida* species.
- ▶ Most common etiology for neonatal meningitis:-**GBS, *E. coli*, and *L. monocytogenes*.**

Others:-S. pneumoniae, other streptococci, non-typable *H.influenzae*, both coagulase-positive and coagulase-negative staphylococci, *Klebsiella,etc*

Table 109-2**Etiologic Agents of Neonatal Pneumonia
According to Timing of Acquisition****TRANSPLACENTAL**

CMV

HSV

*Mycobacterium**Tuberculosis*

Rubella virus

Treponema pallidum

VZV

PERINATAL

Anaerobic bacteria

Chlamydia

CMV

Enteric bacteria

Group B streptococci

Haemophilus influenzae

HSV

*Listeria monocytogenes**Mycoplasma***POSTNATAL**

Adenovirus

*Candida species**Coagulase-negative
staphylococci

CMV

Enteric bacteria*

Enteroviruses

Influenza viruses A, B

Parainfluenza

*Pseudomonas**

RSV

*Staphylococcus aureus**Mycobacterium tuberculosis*

*More likely with mechanical ventilation or indwelling catheters, or after abdominal surgery.



Neonatal sepsis

- **Neonatal sepsis** is a clinical syndrome in an infant 28 days of life or younger, manifested by *systemic signs of infection* and *isolation of a bacterial pathogen* from the bloodstream. A consensus definition for neonatal sepsis is lacking

Classification



- ▶ **Early-onset sepsis..** is defined as the onset of symptoms before **7 days of age.**
 - ▶ is usually due to vertical transmission by ascending contaminated amniotic fluid or during vaginal delivery from bacteria colonizing or infecting the mother's lower genital tract.
- ▶ **Late-onset sepsis....** is defined as the onset of symptoms at **≥ 7 days of age.**
 - ▶ *Maternal vertical transmission*, resulting in initial neonatal colonization that evolves into later infection.
 - ▶ *Horizontal transmission* from direct contact with **care providers** or **environmental sources.**
- ▶ **Late-late onset neonatal sepsis/Very-late-onset sepsis.. (onset after age 1 mo.)**
 - ▶ Source of infection=environment & community



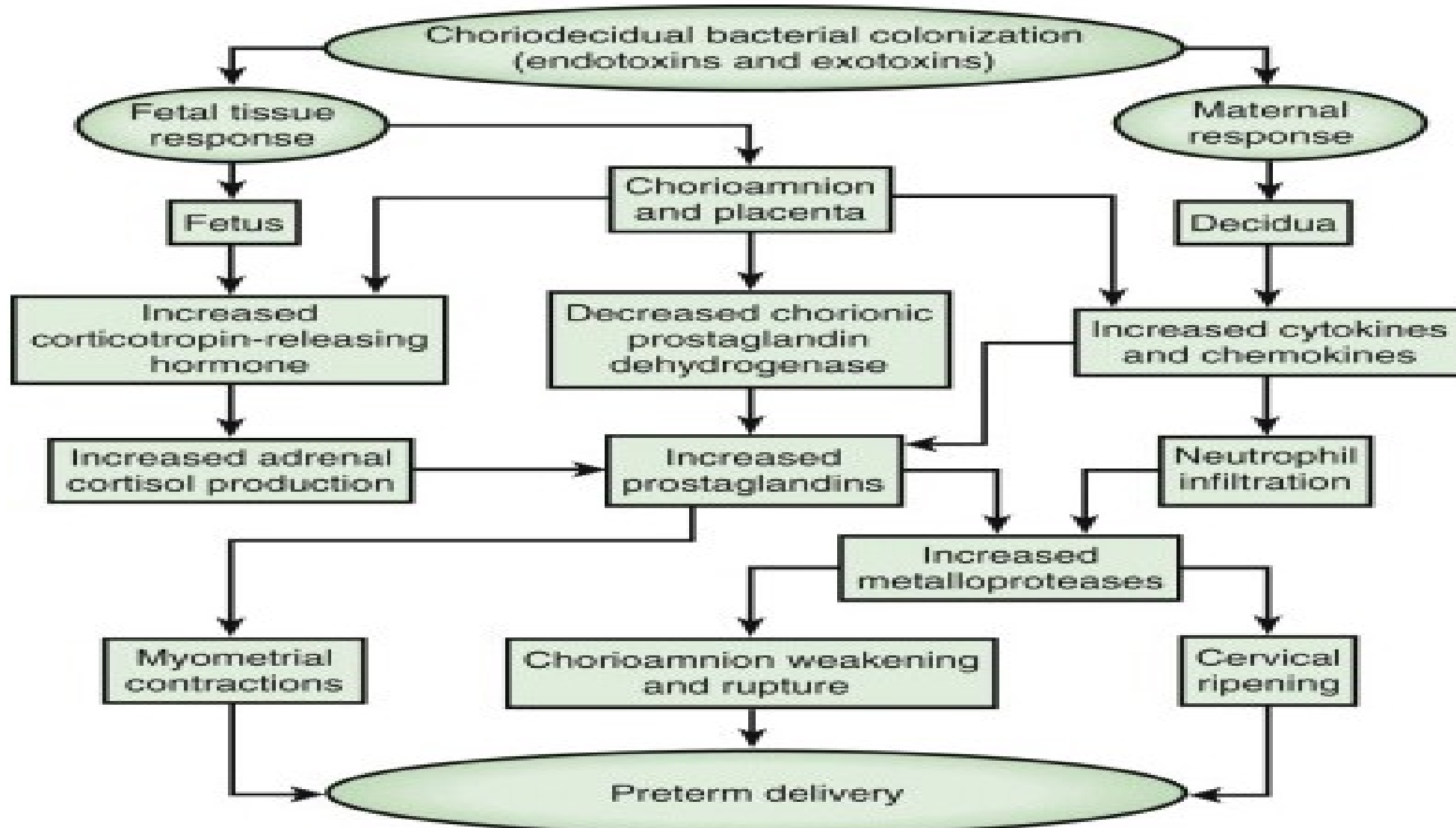
Risk factors

1. Prematurity:-

possible explanation

- a. **maternal genital tract infection is considered to be an important cause of preterm labor.**
- b. the frequency of intra-amniotic infection is *inversely* related to GA.
- c. premature infants have documented **immune dysfunction**.
- d. **prolonged hospitalization** predisposing for nosocomial infection

Potential pathways from chorio-decidual bacterial colonization to preterm delivery





Risk factors...EONS

- ▶ Intrapartum maternal temperature $\geq 38^{\circ}\text{C}$ (100.4°F)
- ▶ Delivery at <37 weeks gestation
- ▶ Chorioamnionitis
- ▶ Five-minute Apgar score ≤ 6
- ▶ Evidence of fetal distress
- ▶ Maternal GBS colonization
- ▶ Membrane rupture ≥ 18 hours





Conti...

- ▶ **Chorioamnionitis** results from microbial invasion of amniotic fluid, but is now better defined by the term **intrauterine inflammation or infection at birth (Triple I)** .
- ▶ **Amniotic infection** may also occur with apparently intact membranes or with a relatively brief duration of membrane rupture.
- ▶ Chorioamnionitis may also be **asymptomatic**, diagnosed only by amniotic fluid analysis or pathologic examination of the placenta.

Table 131.5

Classification of Triple I and Isolated Maternal Fever

TERMINOLOGY FEATURES	
Isolated maternal fever	Maternal oral temperature $\geq 39^{\circ}\text{C}$ is considered a “documented fever.” If the oral temperature is $\geq 38^{\circ}\text{C}$ but $\leq 39^{\circ}\text{C}$, repeat the measurement in 30 min. If the repeat value is $\geq 38^{\circ}\text{C}$, it is considered a “documented fever.”
Suspected Triple I	Fever without a clear source with <i>any</i> of the following:
	1. Baseline fetal tachycardia (>160 beats/min for 10 min) 2. Maternal WBC $>15,000/\text{mm}^3$ 3. Purulent fluid from the cervical os
Confirmed Triple I	All the above (from suspected Triple I) with <i>any</i> of the following:
	1. Amniocentesis-proven infection through positive Gram stain 2. Low glucose of amniotic fluid or positive amniotic fluid culture 3. Placental pathology consistent with infection

Triple I, Intrauterine inflammation or infection at birth; WBC, white blood cell count.

Adapted from Higgins RD, Saade G; Chorioamnionitis Workshop participants: Evaluation and management of women and newborns with a maternal diagnosis of chorioamnionitis: summary of



Conti....

- **Intrapartum antibiotics** are used to reduce vertical transmission of GBS as well as to lessen neonatal morbidity after preterm rupture of membranes.
- Intrapartum chemoprophylaxis **does not reduce the rates of late-onset GBS disease and has no effect on the rates of infection with non-GBS pathogens.**



Late-Onset Infections

- ▶ After birth, neonates are exposed to infectious agents in the neonatal intensive care unit (NICU), the nursery, or in the community (including family).
- ▶ Postnatal infections may be transmitted by direct contact with hospital personnel, the mother, or other family members; from breast milk (HIV, CMV); or from inanimate sources such as contaminated equipment.
- ▶ The most common source of postnatal infections in hospitalized newborns is *hand contamination* of health care personnel, underscoring the importance of handwashing.



Conti...

- ▶ **The incidence of meningitis is 0.2-0.4/1,000 live births** in newborn infants and is higher in preterm infants. Bacterial meningitis may be associated with sepsis or may occur as a local meningeal infection.
- ▶ **Up to one-third of VLBW infants with late-onset meningitis have negative blood culture results.**
- ▶ The discordance between results of blood and cerebrospinal fluid (CSF) cultures suggests that meningitis may be underdiagnosed among VLBW infants and emphasizes the need for culture of CSF in VLBW infants when late-onset sepsis is suspected and in all infants who have positive blood culture results

Bacterial pathogens in neonatal sepsis and focal neonatal infections

	Common pathogens*	Some less common pathogens*
Early onset [¶]		
Term and late preterm infants (GA ≥34 weeks)	■ GBS ■ <i>E. coli</i>	■ <i>Enterobacter</i>, <i>Enterococcus</i>, <i>Klebsiella</i>, <i>Listeria</i>, nontypeable <i>H. influenzae</i>, other enteric gram-negative bacilli, <i>S. aureus</i>, viridans streptococci
Preterm infants (GA <34 weeks)	■ <i>E. coli</i> ■ GBS	■ CoNS, <i>Enterobacter</i>, <i>Klebsiella</i>, <i>Listeria</i>, other enteric and nonenteric gram-negative bacilli, <i>S. aureus</i>, viridans streptococci
Late onset [¶]		
Term and late preterm infants (GA ≥34 weeks)	■ <i>E. coli</i> ■ GBS ■ Additional pathogens seen in the NICU setting – <i>S. aureus</i>, CoNS	■ <i>Enterobacter</i>, <i>Klebsiella</i>, <i>Listeria</i>, <i>N. meningitidis</i>, other enteric and nonenteric gram-negative bacilli, <i>Salmonella</i>, <i>S. pneumoniae</i>, viridans streptococci ■ Additional pathogens seen in the NICU setting – <i>Citrobacter</i>, <i>Enterococcus</i>, <i>Pseudomonas</i>, <i>Serratia</i>
Preterm infants (GA <34 weeks)	■ CoNS ■ <i>S. aureus</i> ■ <i>E. coli</i> ■ <i>Klebsiella</i> ■ GBS	■ <i>Citrobacter</i>, <i>Enterobacter</i>, <i>Enterococcus</i>, <i>Listeria</i>, other enteric and nonenteric gram-negative bacilli, <i>Pseudomonas</i>, <i>Salmonella</i>, <i>Serratia</i>, viridans streptococci
Pathogens based on source of infection		
Meningitis	■ GBS ■ <i>E. coli</i> ■ Other gram-negative enteric bacilli	■ CoNS, <i>Enterococcus</i>, <i>Listeria</i>, <i>N. meningitides</i>, nontypeable <i>H. influenzae</i>, <i>S. aureus</i>, <i>S. pneumoniae</i>, other streptococci (groups A, C, or G and viridans streptococci)
Pneumonia	■ GBS	■ <i>C. trachomatis</i>, <i>Citrobacter</i>, <i>Enterobacter</i>, group A <i>Streptococcus</i>, <i>Klebsiella</i>, <i>Pseudomonas</i>, <i>S. aureus</i>, <i>S. pneumoniae</i>, <i>Serratia</i>
Urinary tract infection ^Δ	■ <i>E. coli</i>	■ <i>Citrobacter</i>, <i>Enterobacter</i>, <i>Enterococcus</i>, <i>Klebsiella</i>, <i>Proteus</i> ■ Additional pathogens seen in the NICU setting – CoNS, <i>S. aureus</i>
Skin and soft tissue infection	■ <i>S. aureus</i> ■ GBS ■ Group A <i>Streptococcus</i>	

Vascular catheter-associated infection	■ <i>S. aureus</i> ■ CoNS ■ <i>Enterococcus</i> ■ Gram-negatives	
Intestinal source/NEC	■ <i>E. coli</i> ■ <i>Klebsiella</i> ■ Other enteric gram-negative bacilli ■ <i>Clostridium</i> spp ■ Anaerobes (eg, <i>Bacteroides</i>)	

GA: gestational age; GBS: group B *Streptococcus*; *E. coli*: *Escherichia coli*; *H. influenzae*: *Haemophilus influenzae*; *S. aureus*: *Staphylococcus aureus*; CoNS: coagulase-negative staphylococci; NICU: neonatal intensive care unit; *N. meningitidis*: *Neisseria meningitidis*; *S. pneumoniae*: *Streptococcus pneumoniae*; *C. trachomatis*: *Chlamydia trachomatis*; NEC: necrotizing enterocolitis.

* This table summarizes bacterial pathogens in neonatal sepsis and in focal neonatal infections. Common pathogens are listed roughly in order of relative frequency within each category; less common pathogens are listed alphabetically. The list of pathogens within each category is not exhaustive. This table does not address nonbacterial causes of neonatal infections (eg, herpes simplex virus, enterovirus, parechovirus, *Candida*). Refer to separate UpToDate content for additional information on nonbacterial pathogens.

¶ The definitions of early and late onset vary in different reports. Within UpToDate content, we generally define early onset as <72 hours after birth and late onset as ≥72 hours. However, some experts use <7 days versus ≥7 days, respectively, for these definitions. The definitions of early and late onset as they pertain to GBS disease are somewhat different. Early-onset GBS infection typically presents within 24 hours of birth but can occur through day 6 after birth. Late-onset GBS typically occurs at 4 to 5 weeks of age.

Δ Urinary tract infections may not be associated with bacteremia in neonates.



Clinical features

- ▶ **May be silent especially in premature newborn**
- ▶ **Symptoms signs are usually non specific**
 - **feeding intolerance & failure to suck**
 - **lethargy and unresponsiveness**
 - **tachypnea and apneic episodes**
 - **grunting and other signs of RD ,etc**
- ▶ **SxSx may vary with time of infection, infecting organism and organ(s) affected.**

Clinical manifestations of bacterial sepsis

GENERAL:-

- ▶ Fever,
- ▶ temperature instability
- ▶ “Not doing well”
- ▶ Poor feeding
- ▶ Edema

GASTROINTESTINAL SYSTEM:-

- ▶ Abdominal distention
- ▶ Vomiting
- ▶ Diarrhea
- ▶ Hepatomegaly

Conti...

RESPIRATORY SYTEM:-

- ▶ Apnea, dyspnea
- ▶ Tachypnea,
- ▶ retractions
- ▶ Flaring, grunting
- ▶ Cyanosis

CARDIOVASCULAR SYSTEM:-

- ▶ Pallor; mottling;
- ▶ cold, clammy skin
- ▶ Tachycardia
- ▶ Hypotension
- ▶ Bradycardia

Conti...

CENTRAL NERVOUS SYSTEM:-

- ▶ Irritability, lethargy
- ▶ Tremors, seizures
- ▶ Hyporeflexia, hypotonia
- ▶ Abnormal Moro reflex
- ▶ Irregular respirations
- ▶ Full fontanel
- ▶ High-pitched cry

HEMATOLOGIC SYSTEM:-

- ▶ Jaundice
- ▶ Splenomegaly
- ▶ Pallor
- ▶ Petechiae, purpura
- ▶ Bleeding



Investigations

- Because of the high mortality rates, there must be a high index of suspicion for neonatal sepsis. Therefore, the following tests, known as septic workup, are done before antibiotics started.



Conti...

1.CBC:-The WBC and ANC are most predictive of infection when these values were low

(WBC < 5,000 and ANC <1,000). ,,,, elevated WBC (>20,000)

=> The combination of **low ANC** and **elevated I/T** ratio (≥ 0.2) is the most predictive combination of WBC indices for EOS.

=>thrombocytopenia & anemia suggests DIC

2.CXR

=>in patients with suspected to have pneumonia

3.Urine analysis & culture

4. Blood culture Gold standard

5.Acute phase reactant





Conti...

6.CSF analysis:-in suspected neonatal meningitis

→ abnormal finding suggesting meningitis

- a. WBC > 30/mm³ with > 60% polymorphs
- b. protein level > 150mg/dl in terms & > 175mg/dl in preterms
- c. CSF glucose < 50% of serum glucose
- d. positive gram stain & culture.

Neonatal Sepsis

CATEGORY PARAMETER		OPTIMAL TIMING, VOLUME OF SPECIMEN, ROUTINE/INVESTIGATIONAL*	APPLICABILITY FOR NEONATAL SEPSIS
CULTURE BASED			
Blood	Culture	>1 mL of whole blood, from 2 sites	Gold standard for bacteremia
CSF	Culture	When clinically feasible	Optimize antimicrobial therapy
Urine	Culture	>72 hr of life	Not useful for EOS; potential benefits for LOS
Tracheal aspirate	Culture	Neonates with endotracheal tube in place and signs of progressive respiratory distress	Usually reflects colonization
NON-CULTURE BASED			
Immune function	MHC II TNF- α	Investigational Investigational	Both decreased in chorioamnionitis and sepsis
Neutrophil indices	Neutropenia Absolute neutrophil count Absolute immature neutrophil count	After 12 hr of life Consider GA, delivery mode, altitude, arterial versus venous sampling, time since birth	Neutropenia better predictor for sepsis than leukocytosis
Neutrophil markers	CD64	Elevated for 24 hr after infection Requires 50 μ L blood Results within hours Investigational	Cut points between 2.38 and 3.62 optimal sensitivity, specificity, and NPV for EOS

Platelet count	Thrombocytopenia and thrombocytosis	Late findings; slow to respond	Thrombocytopenia associated with fungal infection
CSF cell count	CSF WBC	Uninfected neonates: mean 10 cells/mm ³ ; range up to 20 cells/mm ³	Does not predict culture-proven meningitis
CSF chemistries	CSF protein CSF glucose	Term <100 mg/dL Preterm higher; 70–80% of serum glucose	Elevated in fungal meningitis Low glucose specific for bacterial meningitis
Acute phase reactants	CRP Procalcitonin	8–24 hr after infection 2–12 hr after infection	Good NPV Better sensitivity but less specificity than CRP
Sepsis panels/scores		After 24 hr of life Investigational	Most useful for NPV and discontinuation of antimicrobial therapy

* *Investigational* refers to an assay or parameter that is undergoing evaluation for clinical use and applicability.

CRP, C-reactive protein; CSF, cerebrospinal fluid; EOS, early-onset sepsis; GA, gestational age; LOS, late-onset sepsis; MHC II, major histocompatibility complex class II; NPV, negative predictive value; TNF, tumor necrosis factor; WBC, white blood cell count.

From Shane AL, Stoll BJ. Recent developments and current issues in the epidemiology, diagnosis, and management of bacterial and fungal neonatal sepsis, *Am J Perinatol* 30(2):131–141, 2013.



Management of bacterial infections

- ▶ Supportive (O₂, Iv fluid, NGT)
- ▶ Antibiotic-depends on etiology and site of infection
 - ▶ Sepsis without meningitis
 - **Suspected** : Ampicillin(100mg/kg/day + Gentamycin (3-5mg/kg/day) for 7-10 days
 - **G+ve** bacteria: 7-10 days (Im/IV)
 - **G-ve** bacteria: 10-14 days (Im/IV)

Conti...



► Meningitis:

- **G+ve** bacteria: high dose Ampicillin + cephalosporine for **14-21days,IV**
- Double the dose of antibiotics
- **G-ve** bacteria: for *21days (IV)*



GOOD DAY!!



References

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